AP CALCULUS AB

Unit 1 • Lesson 1.15 practice

Lesson 1.15 • Connecting Limits at Infinity and Horizontal Asymptotes

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to \infty} \frac{3x^2 + 1}{x^2 - 5}$ is

(A) 3

(B) $-\frac{1}{5}$

(C) 0

(D) ∞

Question 2

NO CALCULATOR

$\lim_{x \to \infty} \frac{x + 1}{x^2}$ is

(A) 1

(B) 0

(C) ∞

(D) $\frac{1}{2}$

Question 3

NO CALCULATOR

$\lim_{x \to \infty} e^{-x}$ is

(A) 1

(B) ∞

(C) 0

(D) $-\infty$

Question 4

NO CALCULATOR

$\lim_{x \to \infty} \frac{x^2}{e^x}$ is

(A) ∞

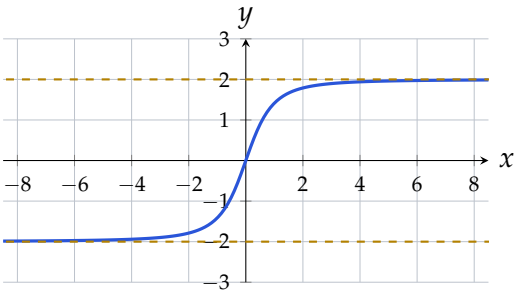
(B) 1

(C) e

(D) 0

Question 5 (graph) NO CALCULATOR

The graph of $f(x) = \dfrac{2x}{\sqrt{x^2+1}}$ is shown with its two horizontal asymptotes. What is $\lim_{x\to-\infty} f(x)$?



(A) 2
(B) 0
(C) -2
(D) $-\infty$

Question 6 (Free response) NO CALCULATOR

Let $f(x) = \dfrac{4x}{\sqrt{9x^2+1}}$.

(a) Find $\lim_{x\to\infty} f(x)$.
(b) Find $\lim_{x\to-\infty} f(x)$. Show why the sign changes.
(c) Write the equations of all horizontal asymptotes of f.
(d) Find $\lim_{x\to\infty} \dfrac{2e^x+5}{e^x-3}$.